LESSON 01 / 10

Where the six ratios come from

For a right triangle with angle θ (acute), opposite O, adjacent A, hypotenuse $H > 0$:

$$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A}$$

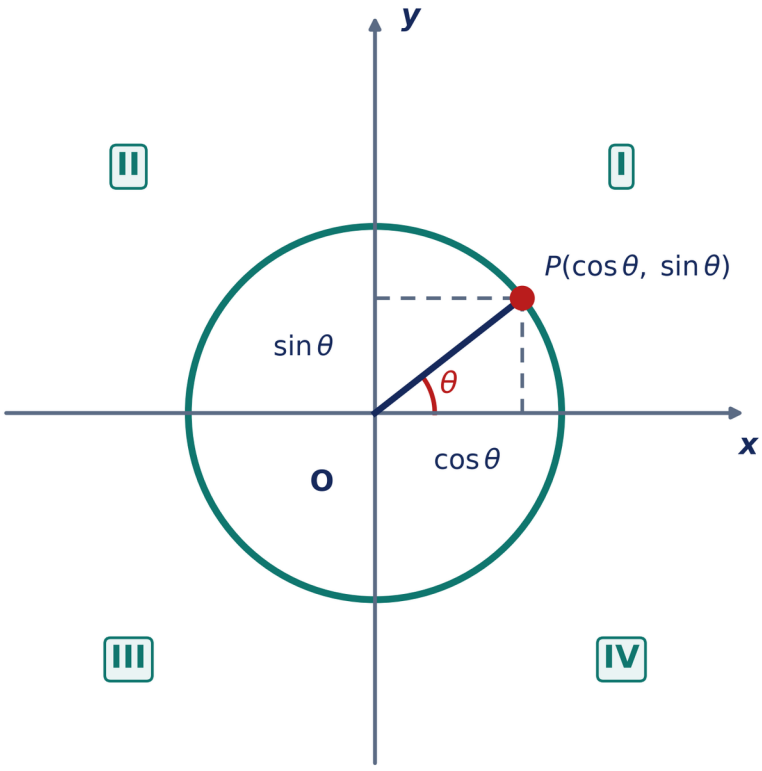
$$\csc\theta = \frac{H}{O}, \quad \sec\theta = \frac{H}{A}, \quad \cot\theta = \frac{A}{O}.$$

Radian convention (the default). An angle is the arc length it cuts on the unit circle: full turn $= 2\pi$ radians $= 360^\circ$, so $180^\circ = \pi$ rad, $90^\circ = \pi/2$, $60^\circ = \pi/3$, $45^\circ = \pi/4$, $30^\circ = \pi/6$. **Never mix degrees and radians in one computation** — pick one system per problem and convert at the boundaries.

The unit circle definition (all angles, not just acute)

The point at angle θ from the positive x-axis on the unit circle is $P = (\cos\theta, \sin\theta)$. This defines $\sin$ and $\cos$ for **all** real θ (negative, $> 2\pi$, whatever), makes them periodic with period 2π, and gives the fundamental identity as the equation of the circle.

The unit circle: P = (cos θ, sin θ) — the definition of all six ratios



$\sin^2\theta + \cos^2\theta = 1$ (every point of the unit circle)

Reciprocal and quotient identities (valid wherever both sides are defined):

$$\sin\theta\csc\theta = 1, \quad \cos\theta\sec\theta = 1, \quad \tan\theta\cot\theta = 1$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

Dividing $\sin^2 + \cos^2 = 1$ *by* $\cos^2$ *or* $\sin^2$ *gives the other two Pythagorean forms:*

$$1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

SHORT DERIVATION

Deriving $1 + \tan^2 = \sec^2$ ***— the pattern that generates all three***

Divide $\sin^2\theta + \cos^2\theta = 1$ *by* $\cos^2\theta$ *(legal when* $\cos\theta \neq 0$*):* $\tan^2\theta + 1 = \sec^2\theta$*. Divide by* $\sin^2\theta$ *(when* $\sin\theta \neq 0$*):* $1 + \cot^2\theta = \csc^2\theta$*. Each Pythagorean identity is one constraint between two coordinates read in a different ratio basis — never memorise three formulas, memorise one and the two divisions.*

ASSUMPTIONS & CONVENTIONS

Angle variable θ *(or* x*) is real; all identities are* ***identities of functions****, true at every angle where both sides are defined. Points of definition:* $\tan, \sec$ *undefined at* $\frac{\pi}{2} + k\pi$ *(cosine zero);* $\cot, \csc$ *undefined at* $k\pi$ *(sine zero),* $k \in \mathbb{Z}$*.* $\sin, \cos$ *have domain* $\mathbb{R}$*, range* $[-1, 1]$*;* $\tan, \cot$ *have range* $\mathbb{R}$*. Units: radians unless a problem states degrees.*

KEY RESULT

Ranges and the inequality toolkit. $-1 \leq \sin\theta, \cos\theta \leq 1$ *for all* θ*; so* $|\sin\theta| \leq 1$*,* $|\cos\theta| \leq 1$*, and expressions like* $3\sin\theta - 4\cos\theta$ *are bounded (*$|a\sin\theta + b\cos\theta| \leq \sqrt{a^2 + b^2}$ *— proved in sheet 3 via compound angles). The reciprocal ratios are unbounded near their poles:* $|\sec| \geq 1$*,* $|\csc| \geq 1$*.*

COMMON TRAP

Common traps — foundations

- $\sin^2\theta + \cos^2\theta = 1$ *does* ***not*** *mean* $\sin\theta + \cos\theta = 1$*; also* $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta = 1 + \sin 2\theta$*.*
- $\sqrt{\sin^2\theta} = |\sin\theta|$*, not* $\sin\theta$*.*
- $\tan\theta = \frac{\sin\theta}{\cos\theta}$ *is undefined wherever* $\cos\theta = 0$ *— never cancel a* $\cos$ *without stating the domain.*
- $\sin(\theta)^{-1} \neq \sin^{-1}\theta$*: the first is* $\csc\theta$*; the second is the inverse function (chapter 4). Write* $\csc$ *and* $\arcsin$ *to stay unambiguous.*

LESSON 02 / 10

Signs — read them off the coordinates

Since $P = (\cos\theta, \sin\theta)$: sine carries the y-sign, cosine the x-sign, tangent their ratio.

Quadrant	sin / csc	cos / sec	tan / cot
I $(0, \frac{\pi}{2})$	+	+	+
II $(\frac{\pi}{2}, \pi)$	+	−	−
III $(\pi, \frac{3\pi}{2})$	−	−	+
IV $(\frac{3\pi}{2}, 2\pi)$	−	+	−

Mnemonic ("**A**ll **S**tudents **T**ake **C**alculus"): I all positive, II only sine/csc, III only tan/cot, IV only cos/sec.

Allied angles — symmetry about the axes

Reflections of $P = (\cos\theta, \sin\theta)$ in the axes give, for example:

$$\sin(\pi - \theta) = \sin\theta, \quad \cos(\pi - \theta) = -\cos\theta$$

$$\sin(\pi + \theta) = -\sin\theta, \quad \cos(\pi + \theta) = -\cos\theta$$

$$\sin(\tfrac{\pi}{2} - \theta) = \cos\theta, \quad \cos(\tfrac{\pi}{2} - \theta) = \sin\theta$$

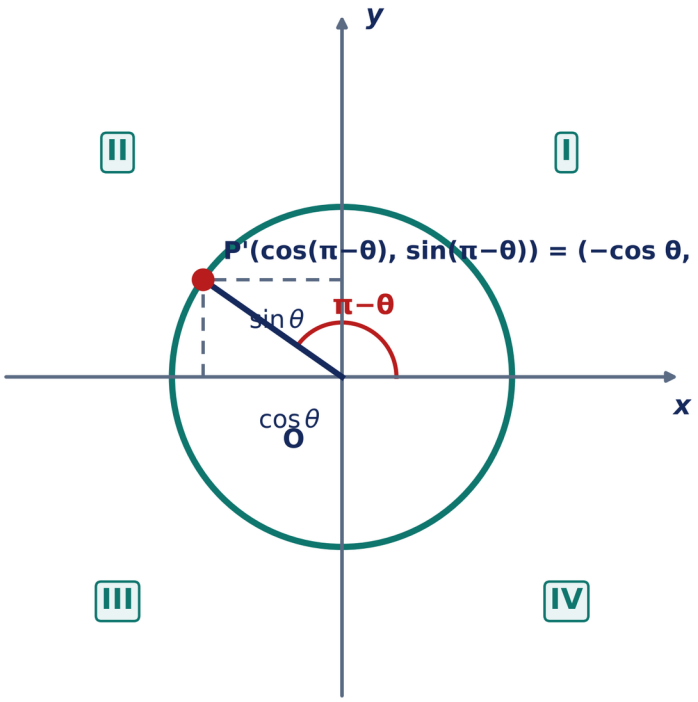
$$\sin(-\theta) = -\sin\theta, \quad \cos(-\theta) = \cos\theta.$$

SHORT DERIVATION

Why odd multiples of $\frac{\pi}{2}$ swap sin and cos

Adding $\frac{\pi}{2}$ rotates P by a quarter turn: $(x, y) \mapsto (-y, x)$. So $\cos(\theta + \frac{\pi}{2}) = -\sin\theta$ and $\sin(\theta + \frac{\pi}{2}) = \cos\theta$ — the coordinates **swap with a sign**. Adding π sends $(x, y) \mapsto (-x, -y)$: both signs flip, no swap. So: **odd multiples of $\frac{\pi}{2}$ swap sine and cosine (with quadrant signs); even multiples only change signs.** This one rotation picture generates the entire allied-angle table.

Allied angle π − θ in QII: sine survives (y unchanged), cosine flips sign



Standard values (memorise the first octant; extend by symmetry)

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	*undef.*

Extend with allied angles: $\sin\frac{2\pi}{3} = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$; $\cos\frac{3\pi}{4} = -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2}$; $\tan\frac{5\pi}{3} = -\tan\frac{\pi}{3} = -\sqrt{3}$. *Special values* $\sin\frac{\pi}{10} = \sin 18^\circ = \frac{\sqrt{5}-1}{4}$ *and* $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$ *are derived in Part II.*

Periodicity: $\sin, \cos$ *have period* 2π; $\tan, \cot$ *have period* π *(because* $\tan(\theta+\pi) = \tan\theta$*; the sine/cosine pair flips sign together and the ratio forgets).* *So* $\sin(\theta + 2k\pi) = \sin\theta$, $\tan(\theta + k\pi) = \tan\theta$.

WORKED EXAMPLE

Worked. Evaluate $\sin 135^\circ$, $\cos 225^\circ$, $\tan\frac{7\pi}{6}$

$\sin 135^\circ = \sin(180^\circ - 45^\circ) = \sin 45^\circ = \frac{\sqrt{2}}{2}$ ✓.

$\cos 225^\circ = \cos(180^\circ + 45^\circ) = -\cos 45^\circ = -\frac{\sqrt{2}}{2}$ ✓. $\tan\frac{7\pi}{6} = \tan\left(\pi + \frac{\pi}{6}\right) = \tan\frac{\pi}{6} = \frac{\sqrt{3}}{3}$ *(QIII: tan positive)* ✓.

COMMON TRAP

Common traps — signs and values

- $\cos(\pi - \theta) = -\cos\theta$ **not** $\cos\theta$; the cosine sign flips in QII, the sine does not.
- $\sin^{-1}$ notation trap again: $\sin(\pi - \theta)$ is an allied angle, not an inverse function.
- $\tan\frac{\pi}{2}$ is **undefined** (not ∞ as a number); "approaches $+\infty$ from the left" is a limit statement.
- Degree/radian mixing: $135^\circ = \frac{3\pi}{4}$ rad — write the conversion before computing.

LESSON 03 / 10

The formulae

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B,$$

$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}.$$

Sign reading: ***sine keeps the sign of ±; cosine flips it*** *($\cos(A-B)$ has a +). Tangent's denominator flips like cosine's.*

SHORT DERIVATION

Derivation of $\cos(A-B)$ from the unit circle (the root of all four)

Points $P=(\cos A,\sin A)$, $Q=(\cos B,\sin B)$ on the unit circle. The chord $|PQ|^2 = 2-2\cos(A-B)$ (law of cosines in $\triangle POQ$ with angle $A-B$ at the centre; or expand $(\cos A-\cos B)^2+(\sin A-\sin B)^2$ directly). Expanding: $|PQ|^2 = 2-2(\cos A\cos B+\sin A\sin B)$. Comparing: $\cos(A-B)=\cos A\cos B+\sin A\sin B$ ✓. Replace $B\to -B$ (cos even, sin odd) for $\cos(A+B)$; use $\sin x=\cos(\frac{\pi}{2}-x)$ to get both sine formulae; divide sine by cosine for tangent. All four identities rest on one distance computation.

Standard applications

Evaluate non-standard angles.
$\sin 75^\circ = \sin(45^\circ+30^\circ) = \frac{\sqrt2}{2}\frac{\sqrt3}{2}+\frac{\sqrt2}{2}\frac12 = \frac{\sqrt6+\sqrt2}{4} \approx 0.9659$ ✓.
$\tan 105^\circ = \tan(60^\circ+45^\circ) = \frac{\sqrt3+1}{1-\sqrt3} = -(2+\sqrt3) \approx -3.732$ ✓ *(rationalise the denominator).*

Given one ratio and a quadrant, find the rest, then expand. *If $\sin A=\frac35$, A acute, then $\cos A=\frac45$; if $\cos B=\frac{5}{13}$, B acute, then $\sin B=\frac{12}{13}$, and $\sin(A+B)=\frac35\frac{5}{13}+\frac45\frac{12}{13}=\frac{63}{65}$ ✓.*

The linear form (preview of equations sheet 7). $a\sin x+b\cos x=\sqrt{a^2+b^2}\sin(x+\varphi)$ *where $\cos\varphi=\frac{a}{\sqrt{a^2+b^2}}$, $\sin\varphi=\frac{b}{\sqrt{a^2+b^2}}$ (so $\tan\varphi=b/a$ with the quadrant fixed by the two signs). Consequence (range): $|a\sin x+b\cos x|\le\sqrt{a^2+b^2}$ — equality at $x+\varphi=\pm\frac{\pi}{2}$.*

KEY RESULT

Pattern ownership. *Every compound-angle exercise is one of four moves: (1) expand and collect like terms; (2) factor a collected expression back into a product of sines/cosines (using the formulae in reverse); (3) plug special angles $15^\circ, 75^\circ, 105^\circ$ etc.; (4) given $\sin A$ + quadrant data, rebuild $\cos A, \tan A$ with* ***Pythagoras plus a sign read from the quadrant****, then expand. Doing (4) with a wrong sign is the most common exam loss.*

WORKED EXAMPLE

Worked. Prove $\cos(A-B)-\cos(A+B)=2\sin A\sin B$ **and evaluate** $\cos 75^\circ-\cos 15^\circ$

Expand both:

$\cos(A-B)-\cos(A+B)=(\cos A\cos B+\sin A\sin B)-(\cos A\cos B-\sin A\sin B)=2s$

✓. *With* $A=75^\circ, B=15^\circ$: *sum-to-product gives*

$-2\sin 45^\circ\sin 30^\circ=-2\cdot\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=-\frac{\sqrt{2}}{2}\approx -0.7071$ ✓ *(matches direct:* $0.2588-0.9659$*).*

COMMON TRAP

Common traps — compound angles

- $\cos(A-B)=\cos A\cos B+\sin A\sin B$ — *the plus sign. #1 exam error in this chapter.*
- $\tan(A+B)$ *denominator is* $1-\tan A\tan B$; *numerator keeps* $+$. *Mixing these gives* $\tan 45^\circ\neq 1$ *instantly.*
- *Formulae need the angles in a* ***common unit***: *convert* $A=20^\circ$, $B=\frac{\pi}{6}$ *before expanding.*
- $\sin(A+B)\neq\sin A+\sin B$; $\cos 2A\neq 2\cos A$ — *the formulae are products, not linear.*
- *Finding* $\cos A$ *from* $\sin A=\frac{3}{5}$ *gives* $\pm\frac{4}{5}$; *the quadrant decides. Acute* $\Rightarrow +$. *State the quadrant assumption if given.*

LESSON 04 / 10

Double angles (set $B = A$ in the compound formulae)

$$\sin 2A = 2\sin A\cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}.$$

The three forms of $\cos 2A$ are one identity seen through $\sin^2 + \cos^2 = 1$; each is the natural form for a different substitution (in terms of $\cos$ alone, $\sin$ alone, or mixed).

Power reduction (inverse reading — used constantly):

$$\sin^2 A = \frac{1 - \cos 2A}{2}, \qquad \cos^2 A = \frac{1 + \cos 2A}{2}$$

$$\sin A\cos A = \frac{\sin 2A}{2}$$

Triple angles (set $B = 2A$, expand)

$$\sin 3A = 3\sin A - 4\sin^3 A$$

$$\cos 3A = 4\cos^3 A - 3\cos A.$$

SHORT DERIVATION

Derivation — ten seconds, no memorising

$\sin 3A = \sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A$ ✓. *The cosine version is the same computation with the cosine formulae. Inverting them:* $3\sin A - 4\sin^3 A = \sin 3A$ *solves cubics of that shape — the standard move behind* $\sin 18^\circ$*-style values (Part II).*

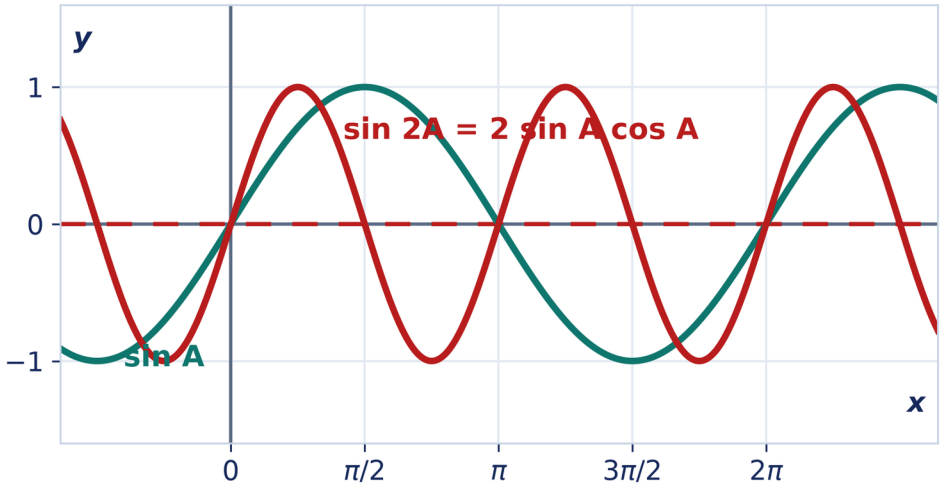
Half angles (solve the double-angle forms for $A/2$)

$$\sin\frac{A}{2} = \pm\sqrt{\frac{1 - \cos A}{2}}, \qquad \cos\frac{A}{2} = \pm\sqrt{\frac{1 + \cos A}{2}}$$

$$\tan\frac{A}{2} = \frac{1 - \cos A}{\sin A} = \frac{\sin A}{1 + \cos A} = \pm\sqrt{\frac{1 - \cos A}{1 + \cos A}}.$$

The two rational forms of $\tan\frac{A}{2}$ carry ***no square root*** *— prefer them whenever $\tan$ is all you need. The $\pm$ in the radical forms is resolved from the* ***quadrant of*** *$\frac{A}{2}$, not of A.*

sin A vs sin 2A: the double-angle curve oscillates twice as fast



KEY RESULT

Forms to convert between, fluently: $\sin 2A = 2\sin A\cos A$; $\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$; *power reduction* $\sin^2, \cos^2 \leftrightarrow \cos 2A$; *products* $\sin A\cos A \leftrightarrow \sin 2A$. *Converting an equation into* $\sin 2A$*-language often halves its degree — that is the point of the machinery.*

WORKED EXAMPLE

Worked. If $\cos A = \frac{3}{5}$, A **acute, find** $\cos 2A$, $\sin 2A$, $\tan\frac{A}{2}$. $\sin A = \frac{4}{5}$ *(acute* $\Rightarrow +$*).* $\cos 2A = 2\cdot\frac{9}{25} - 1 = -\frac{7}{25}$ ✓. $\sin 2A = 2\cdot\frac{4}{5}\cdot\frac{3}{5} = \frac{24}{25}$ ✓ *(check:* $(-\frac{7}{25})^2 + (\frac{24}{25})^2 = \frac{49+576}{625} = 1$ ✓*).* $\tan\frac{A}{2} = \frac{1-\cos A}{\sin A} = \frac{2/5}{4/5} = \frac{1}{2}$ ✓ *(positive:* $A/2$ *acute).*

COMMON TRAP

Common traps — multiple angles

- *Half-angle ±: for* $A = 60^\circ$, $\frac{A}{2} = 30^\circ$ *(QI)* $\Rightarrow +$*; for* $A = 420^\circ$, $\frac{A}{2} = 210^\circ$ *(QIII)* $\Rightarrow$ *sine root negative. Decide from* $\frac{A}{2}$*'s quadrant.*
- $\cos 2A = 1 - 2\sin^2 A$ *solved for* $\sin A$ *gives* **two** *roots* $(\pm)$*; the quadrant of* A *selects.*
- $\tan 2A$ *is undefined when* $1 - \tan^2 A = 0$*, i.e.* $A = \frac{\pi}{4} + k\frac{\pi}{2}$ *— consistent with* $\tan 2A$ *blowing up at* $2A = \frac{\pi}{2} + k\pi$.
- $(\sin A \pm \cos A)^2 = 1 \pm \sin 2A$ *— remember the middle term.*

LESSON 05 / 10

The four product-to-sum formulae

Add/subtract the compound sine and cosine expansions:

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

SHORT DERIVATION

Derivation — one addition, one subtraction
$\sin(A+B) = \sin A\cos B + \cos A\sin B$ *and* $\sin(A-B) = \sin A\cos B - \cos A\sin B$. *Add:* $2\sin A\cos B = \sin(A+B) + \sin(A-B)$. *Subtract:* $2\cos A\sin B = \sin(A+B) - \sin(A-B)$. *Same game with the cosine pair.* ***The RHS argument pair is always*** $(A+B, A-B)$ *— one formula in each direction, nothing new to memorise.*

Sum-to-product (set $C = A+B$, $D = A-B$, so $A = \frac{C+D}{2}$, $B = \frac{C-D}{2}$)

$$\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$$

$$\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$$

Note the ***minus sign*** *in the last one (the only formula with it).*

KEY RESULT

What these are for. *(1) Turn products of sines/cosines into sums (then factor); (2) turn sums into products (then solve each factor* $= 0$ *— the backbone of equation sheet 7); (3) telescope sums whose arguments are in arithmetic progression (Part II series); (4) evaluate exact values:*
$\sin 75^\circ + \sin 15^\circ = 2\sin 45^\circ\cos 30^\circ = 2\cdot\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{2} \approx 1.2247$ ✓ *matches direct addition* $0.9659 + 0.2588$.

WORKED EXAMPLE

Worked. Simplify $\cos 3x\cos x + \cos 5x\cos x$-type products; evaluate $\cos 75^\circ - \cos 15^\circ$

Product-to-sum: $\cos 3x\cos x = \frac{1}{2}[\cos 4x + \cos 2x]$; *so the sum* $= \frac{1}{2}[\cos 4x + \cos 2x + \cos 6x + \cos 4x] = \frac{1}{2}[\cos 2x + 2\cos 4x + \cos 6x]$. *For the exact value:* $\cos 75^\circ - \cos 15^\circ = -2\sin 45^\circ \sin 30^\circ = -\frac{\sqrt{2}}{2} \approx -0.7071$ ✓ *(direct:* $0.2588 - 0.9659$*).*

WORKED EXAMPLE

Worked. Factor $\sin x + \sin 3x + \sin 5x$

Pair first and last? Pairwise sum-to-product: $\sin x + \sin 5x = 2\sin 3x\cos 2x$, *so the expression* $= 2\sin 3x\cos 2x + \sin 3x = \sin 3x(2\cos 2x + 1)$. *Factored form ready for solving* $\sin 3x(2\cos 2x + 1) = 0$ *(sheet 7's method). Check at* $x = 30^\circ$*:* $\sin 30^\circ + \sin 90^\circ + \sin 150^\circ = 0.5 + 1 + 0.5 = 2$; *RHS:* $\sin 90^\circ(2\cos 60^\circ + 1) = 1 \cdot (1 + 1) = 2$ ✓.

COMMON TRAP

Common traps — transformations

- $2\sin A\sin B = \cos(A - B) - \cos(A + B)$: *the* **order** *of* $A - B$, $A + B$ *matters (sign).*
- $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$ *— the leading minus.*
- *Mixed arguments:* $2\sin A\cos B$ *vs* $2\cos A\sin B$ *are different formulas; match the pattern before applying.*
- *Keep the* $\frac{C \pm D}{2}$ *halving straight:* $\sin C + \sin D$ *uses* $\sin\frac{C+D}{2}\cos\frac{C-D}{2}$ *(sine of the average, cosine of the half-difference).*

LESSON 06 / 10

The idea: one circle, infinitely many angles

A trig equation's solutions repeat every period. The **principal solution** lies in $[0, 2\pi)$ (for $\sin, \cos$) or $[0, \pi)$ (for $\tan$); the **general solution** packages all copies:

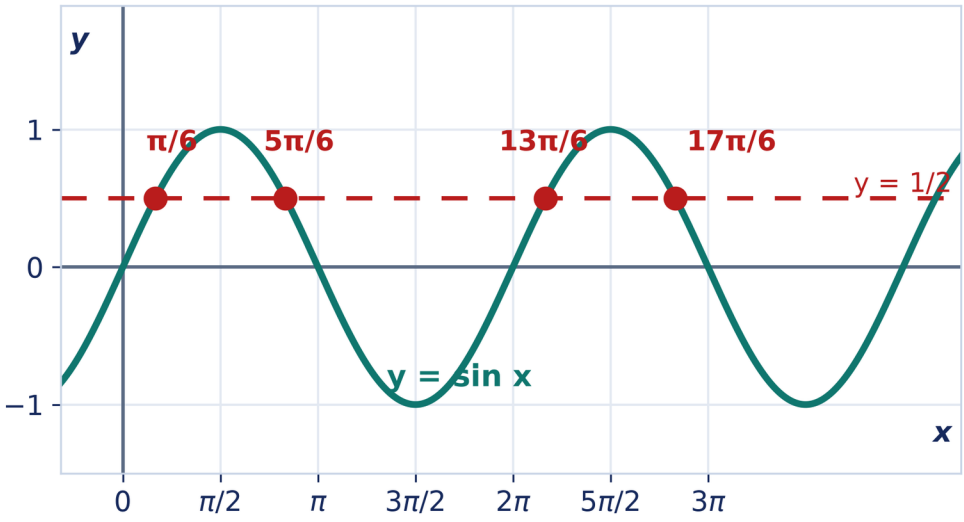
Equation	General solution	Why
$\sin\theta = \sin\alpha$	$\theta = n\pi + (-1)^n\alpha$, $n \in \mathbb{Z}$	sin is α at α and $\pi - \alpha$ per period 2π
$\cos\theta = \cos\alpha$	$\theta = 2n\pi \pm \alpha$	cos is α at $\pm\alpha$ per period 2π
$\tan\theta = \tan\alpha$	$\theta = n\pi + \alpha$	tan has period π

SHORT DERIVATION

Why $(-1)^n$ appears in the sine solution

Inside $[0, 2\pi)$ the solutions of $\sin\theta = \sin\alpha$ are α and $\pi - \alpha$ (two symmetric points on the horizontal line $y = \sin\alpha$). Adding 2π-copies: $\alpha + 2n\pi$ and $\pi - \alpha + 2n\pi = (2n+1)\pi - \alpha$. The single formula $n\pi + (-1)^n\alpha$ interleaves both families: even n gives $2k\pi + \alpha$, odd n gives $(2k+1)\pi - \alpha$ ✓. For cosine the two solutions $\pm\alpha$ are symmetric about 0, giving $2n\pi \pm \alpha$. For tangent the period is π and there is one solution per period.

sin x = 1/2: two solutions per period — x = nπ + (−1)^n π/6



The discipline of solving

1. **Reduce** to one function where possible ($\sin^2$-type: use Pythagorean or power reduction).
2. **Find principal solutions** on one period (draw the circle or the wave).
3. **Write the general form** using the table (substitute the α you found).
4. **Intersect with any requested interval** by ranging over integer n.
5. **Check** solutions in the original equation if any step squared, divided, or used a derived identity.

WORKED EXAMPLE

Worked. Solve $\sin x = \frac{1}{2}$ generally; count solutions in $[0, 2\pi)$

Principal: $x = \frac{\pi}{6}, \frac{5\pi}{6}$. General: $x = n\pi + (-1)^n \frac{\pi}{6}$ ✓ (check $n = 2$: $2\pi + \frac{\pi}{6} = \frac{13\pi}{6}$, $\sin = \frac{1}{2}$ ✓; $n = 3$: $3\pi - \frac{\pi}{6} = \frac{17\pi}{6}$ ✓). In $[0, 2\pi)$: exactly **2** solutions.

WORKED EXAMPLE

Worked. Solve $\cos x = \frac{1}{2}$ and $\tan x = 1$ generally

$\cos x = \cos \frac{\pi}{3} \Rightarrow x = 2n\pi \pm \frac{\pi}{3}$. $\tan x = \tan \frac{\pi}{4} \Rightarrow x = n\pi + \frac{\pi}{4}$. Checks: $\cos(\pm\frac{\pi}{3} + 2\pi) = \frac{1}{2}$ ✓; $\tan(\frac{\pi}{4} + \pi) = \tan \frac{5\pi}{4} = 1$ ✓.

COMMON TRAP

Common traps — general solutions

- Forgetting $(-1)^n$: $\theta = n\pi + \alpha$ is the **tangent** form; for sine it drops half the solutions ($\sin(\pi + \alpha) = -\sin\alpha \neq \sin\alpha$).
- $\cos x = c$ with $|c| > 1$: **no solution**; $|c| = 1$: $x = 2n\pi$ (or $2n\pi + \pi$); state this before "solving".
- Mixing units inside one formula: $x = n\pi + (-1)^n 30^\circ$ is illegal — either all radians ($x = n\pi + (-1)^n \frac{\pi}{6}$) or all degrees ($x = n \cdot 180^\circ + (-1)^n 30^\circ$).
- $\tan x$ general solution adds π, not 2π; period of $\tan, \cot$ is π.
- Squared equations ($\sin^2 x = \frac{1}{4}$) give $\sin x = \pm\frac{1}{2}$ — four principal solutions per 2π; do not drop the $\pm$.

LESSON 07 / 10

Method 1 — factorise, never divide

Bring everything to one side and factor. Solve each factor with the master forms.

WORKED EXAMPLE

Worked. Solve $\sin 2x = \cos x$

$2\sin x\cos x = \cos x \Rightarrow \cos x(2\sin x - 1) = 0 \Rightarrow \cos x = 0$ **or** $\sin x = \frac{1}{2}$. So $x = \frac{\pi}{2} + n\pi$ or $x = n\pi + (-1)^n\frac{\pi}{6}$. In $[0, 2\pi)$: $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ — four solutions ✓ (each verifies in $\sin 2x = \cos x$). **Dividing by** $\cos x$ **would have destroyed the** $\frac{\pi}{2}, \frac{3\pi}{2}$ **family** — the classic trap.

COMMON TRAP

Golden rule — never divide by a trig expression

$\cos x$ vanishes somewhere, so cancelling it loses solutions. Multiply out, bring over, factor. Same for $\sin x$, $\tan x$, and for any substitution step — check what it excludes.

Method 2 — the harmonic form $a\sin x + b\cos x = c$

Write $a\sin x + b\cos x = R\sin(x + \varphi)$ with $R = \sqrt{a^2 + b^2}$ and $\tan\varphi = b/a$ (quadrant of φ fixed by signs of a, b). Then $\sin(x + \varphi) = c/R$, which needs $|c| \le R$ — the **compatibility test** (no solution if $|c| > R$; one per period if $|c| = R$; two per period if $|c| < R$).

SHORT DERIVATION

Worked. Solve $\sqrt{3}\sin x + \cos x = 1$

$R = \sqrt{3+1} = 2$; $\tan\varphi = \frac{1}{\sqrt{3}}$, $\cos\varphi = \frac{\sqrt{3}}{2}$, $\sin\varphi = \frac{1}{2} \Rightarrow \varphi = \frac{\pi}{6}$. So $2\sin(x + \frac{\pi}{6}) = 1 \Rightarrow \sin(x + \frac{\pi}{6}) = \frac{1}{2} \Rightarrow x + \frac{\pi}{6} = n\pi + (-1)^n\frac{\pi}{6}$. Even $n = 2k$: $x = 2k\pi$; odd $n = 2k+1$: $x = 2k\pi + \frac{2\pi}{3}$. In $[0, 2\pi)$: $x = 0, \frac{2\pi}{3}$ ✓ (check $x = 0$: $0 + 1 = 1$ ✓; $x = \frac{2\pi}{3}$: $\sqrt{3}\cdot\frac{\sqrt{3}}{2} - \frac{1}{2} = 1$ ✓).

Method 3 — polynomial in one ratio

$2\cos^2 x + 3\sin x = 0 \Rightarrow 2(1 - \sin^2 x) + 3\sin x = 0 \Rightarrow 2\sin^2 x - 3\sin x - 2 = 0 \Rightarrow (2\sin x + 1$

Only $\sin x = -\frac{1}{2}$ is admissible ($\sin x = 2$ is impossible — range!) $\Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$ per period.

Always clip roots of the polynomial to $[-1, 1]$ for sin/cos.

Method 4 — squaring (last resort) and filtering

$\sin x + \cos x = 1$: square $\Rightarrow 1 + \sin 2x = 1 \Rightarrow \sin 2x = 0 \Rightarrow x = \frac{n\pi}{2}$. Squaring **added** roots: check each in the original: $x = 0$ ✓, $x = \frac{\pi}{2}$ ✓, $x = \pi$ gives $-1 \ne 1$ ✗, $x = \frac{3\pi}{2}$ gives $-1 \ne 1$ ✗. Final: $x = 2n\pi$ or $x = \frac{\pi}{2} + 2n\pi$. (Better: rewrite as $\sqrt{2}\sin(x + \frac{\pi}{4}) = 1$ — Method 2 avoids ghosts entirely.)

Interval filtering: solutions of √3 sin x + cos x = 1 in [0, 2π) — the general form must be ranged over n



KEY RESULT

Method chooser. *(1) Can I factor? Factor (Method 1). (2) Is it* $a\sin x + b\cos x$*? Harmonic form (Method 2 — it also tests solvability). (3) Is it quadratic in one ratio? Substitute and clip to range (Method 3). (4) Absolute values? Split cases per interval of the inside expression. (5) Only if desperate: square, then filter (Method 4).*

COMMON TRAP

Traps — equations II

- $\sin 2x = \cos x$*-type: convert to one angle multiple only after moving everything to one side.*
- *Polynomial in* $\sin x$*: reject* $|\sin x| > 1$ *roots* ***before*** *writing answers.*
- *Squaring: every step after it is suspect; verify all candidates in the original.*
- *Counting problems ("how many roots in* $[a, b]$*?"): draw the wave, count crossings — then convert the general solution with the right integer range for* n*.*
- $\tan x = \tan y \Rightarrow x = y + n\pi$ *only; it is* ***not*** $x = y + 2n\pi$*.*

LESSON 08 / 10

The model: one constraint, many identities

If $A+B+C=\pi$ (angles of a triangle, or any three reals summing to π), then $C=\pi-(A+B)$, so $\sin C=\sin(A+B)$, $\cos C=-\cos(A+B)$, $\tan C=-\tan(A+B)$. Substituting the compound formulae collapses symmetric expressions. **This is derived from Part I machinery — no new theory, just the constraint exploited.**

| Identity (given $A+B+C=\pi$) | Form | |---|---| | Tangents |
$\tan A+\tan B+\tan C=\tan A\tan B\tan C$ | | Sine double angles |
$\sin 2A+\sin 2B+\sin 2C=4\sin A\sin B\sin C$ | | Cosines squared |
$\cos^2 A+\cos^2 B+\cos^2 C+2\cos A\cos B\cos C=1$ | | Sines |
$\sin A+\sin B+\sin C=4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$ | | Half-angle tangents |
$\tan\frac{A}{2}\tan\frac{B}{2}+\tan\frac{B}{2}\tan\frac{C}{2}+\tan\frac{C}{2}\tan\frac{A}{2}=1$ |

SHORT DERIVATION

Derive the tangent identity (the exam workhorse)
$\tan(A+B)=\tan(\pi-C)=-\tan C$, and $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$. So $\tan A+\tan B=-\tan C(1-\tan A\tan B)$, i.e. $\tan A+\tan B+\tan C=\tan A\tan B\tan C$ ✓. (Fails exactly when some $\cos$ vanishes — a 90° angle — then read the identity in the limit or in the cotangent form.)

SHORT DERIVATION

Derive $\sin 2A+\sin 2B+\sin 2C=4\sin A\sin B\sin C$ (model derivation)
$\sin 2A+\sin 2B=2\sin(A+B)\cos(A-B)$ and $\sin 2C=\sin(2\pi-2(A+B))=-\sin 2(A+B)=-2\sin(A+B)\cos(A+B)$. Sum: $2\sin(A+B)[\cos(A-B)-\cos(A+B)]=2\sin(A+B)\cdot 2\sin A\sin B=4\sin A\sin B\sin C$ (using $\sin(A+B)=\sin C$ and the sum-to-product identity from sheet 5) ✓. Numeric check $(60^\circ,60^\circ,60^\circ)$: LHS $=3\sin 120^\circ=2.598=$ RHS $=4(\frac{\sqrt{3}}{2})^3=2.598$ ✓; $(30^\circ,60^\circ,90^\circ)$: both 1.732 ✓.

Applications

Symmetric tangent expressions. If A,B,C are angles of a triangle and $\tan A=1$, $\tan B=2$, then $\tan C=-\tan(A+B)=-\frac{1+2}{1-2}=3$; check the identity: $1+2+3=6=1\cdot 2\cdot 3$ ✓.

WORKED EXAMPLE

Worked (classic). Evaluate $\tan 20^\circ \tan 40^\circ \tan 80^\circ$

Use $\tan 3\theta = \frac{3t-t^3}{1-3t^2}$ *at* $\theta = 20^\circ$ *where* $\tan 60^\circ = \sqrt{3}$; $t = \tan 20^\circ$ *satisfies* $t^3 - 3\sqrt{3}\,t^2 - 3t + \sqrt{3} = 0$. *The roots are* $\tan 20^\circ, \tan 80^\circ, \tan 140^\circ = -\tan 40^\circ$ *(adding* $60^\circ, 120^\circ$ *preserves* $\tan 3\theta$*). Product of roots* $= -\sqrt{3}$, *so* $\tan 20^\circ \tan 80^\circ(-\tan 40^\circ) = -\sqrt{3} \Rightarrow \tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3} \approx 1.7321$ ✓ *(direct:* $0.36397 \times 0.83910 \times 5.67128 = 1.7321$*).*

KEY RESULT

When to use conditional identities. *Spot the hidden constraint: "angles of a triangle",* $A+B+C=\pi$ *(or* $=180^\circ$, $=2\pi -$ *something), or complementary/linked angles like* $A+B=\frac{\pi}{2}$. *Then substitute the constraint into one angle and let the identity do the algebra. For products like* $\tan 20 \tan 40 \tan 80$ *the* ***triple-angle polynomial*** *(roots-of-Vieta) is the standard second tool — Part II generalises it to roots-of-unity products.*

COMMON TRAP

Traps — conditional identities

- *The identity* $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ *needs all tangents defined; a right angle breaks it (one side undefined).*
- *Degrees vs radians inside one identity — convert before substituting* ($180^\circ = \pi$).
- *The constraint must actually hold: "*A, B, C *are three angles" without "of a triangle" does* ***not*** *give* $A+B+C=\pi$.
- *Vieta on the triple-angle cubic gives* $\sum$ *of roots* $= 3\sqrt{3}$ *and product* $= -\sqrt{3}$; *track the sign flip from* $\tan 140^\circ = -\tan 40^\circ$ *carefully — it is where errors creep in.*

LESSON 09 / 10

The whole toolkit on one sheet

Definitions. Unit circle: $P = (\cos\theta, \sin\theta)$; $\tan = \sin/\cos$; $\sin^2 + \cos^2 = 1$; $1 + \tan^2 = \sec^2$; $1 + \cot^2 = \csc^2$ (divide the master identity). Radians default: $\pi = 180^\circ$. Ranges: $\sin, \cos \in [-1, 1]$; periods: $\sin, \cos$ are 2π, $\tan, \cot$ are π.

Signs & allied angles. Quadrant signs (ASTC); odd multiples of $\frac{\pi}{2}$ swap $\sin \leftrightarrow \cos$ with quadrant signs, even multiples only sign; $\sin(\pi - \theta) = \sin\theta$, $\cos(\pi - \theta) = -\cos\theta$.

Compound angles. $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$; $\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$ (flip!); $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$. Derived bound: $|a\sin x + b\cos x| \le \sqrt{a^2 + b^2}$.

Multiple/submultiple. $\sin 2A = 2\sin A\cos A$; $\cos 2A$ three forms; power reduction $\sin^2 = \frac{1 - \cos 2A}{2}$, $\cos^2 = \frac{1 + \cos 2A}{2}$; $\sin 3A = 3\sin A - 4\sin^3 A$, $\cos 3A = 4\cos^3 A - 3\cos A$; half-angle radicals with $\pm$ from the quadrant of $\frac{A}{2}$; $\tan\frac{A}{2} = \frac{1 - \cos A}{\sin A} = \frac{\sin A}{1 + \cos A}$ (no radical form needed).

Transformations. $2\sin A\cos B = \sin(A + B) + \sin(A - B)$ etc. (4 formulas); sum-to-product with the minus sign in $\cos C - \cos D = -2\sin\frac{C + D}{2}\sin\frac{C - D}{2}$.

Equations. Master general solutions: $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\alpha$; $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$; $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$. Methods: factorise (never divide), harmonic form $R\sin(x + \varphi)$ with compatibility $|c| \le R$, polynomial in one ratio with range clip, squaring + filtering, interval counting by ranging n.

Conditional ($A + B + C = \pi$). $\tan A + \tan B + \tan C = \tan A\tan B\tan C$; $\sin 2A + \sin 2B + \sin 2C = 4\sin A\sin B\sin C$; half-angle product sum $= 1$.

KEY RESULT

Five questions to ask any trig problem. (1) Degrees or radians? (2) Which quadrant is the angle in (signs)? (3) Am I allowed to divide/cancel (any factor that can vanish)? (4) After any squaring/substitution, which solutions are ghosts? (5) Does a hidden constraint ($A + B + C = \pi$, complementary angles) trigger a conditional identity?

ASSUMPTIONS & CONVENTIONS

All angles real; radians unless stated; identities hold wherever both sides are defined; general solutions are over $n \in \mathbb{Z}$; square roots in half-angle formulas carry the sign of the quadrant of $\frac{A}{2}$; "solve generally" means the general solution, "solve in $[a, b)$" means filter with integer n.

STANDARD APPLICATION

Where this reappears. Compound-angle machinery powers the calculus of sin/cos (derivatives), vectors (angles, projections), 2D/3D geometry (direction angles), complex numbers (de Moivre), and periodic modelling. Equation discipline (domain, factorise, filter) is exactly the discipline of solving with substitutions in calculus and of Diophantine angle problems in olympiads (Part II).

COMMON TRAP

Final trap sweep

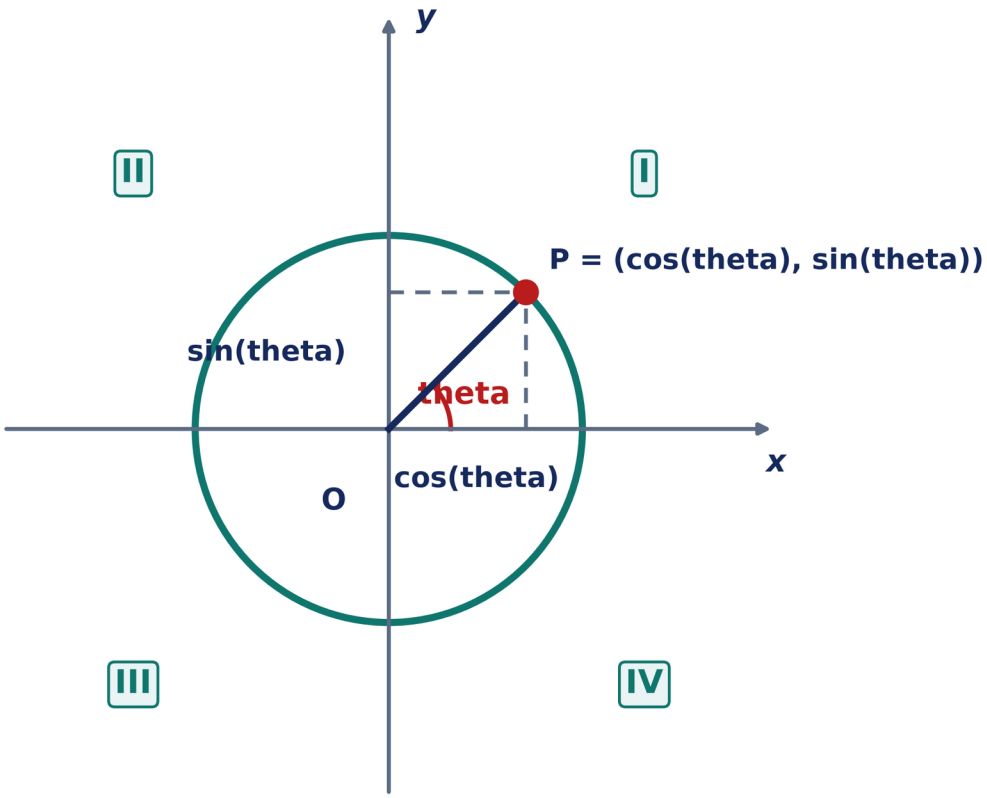
- $\cos(A-B) = \cos A\cos B + \sin A\sin B$ — plus sign, every time.
- $\theta = n\pi + (-1)^n\alpha$ for sine (not $n\pi + \alpha$).
- Divide by $\cos x$? Don't. Factor instead.
- $\sqrt{\sin^2 x} = |\sin x|$; half-angle radicals need the quadrant of $\frac{A}{2}$.
- $\sin 2x = \sin 2y \Rightarrow x = y + k\pi$ **or** $x = \frac{\pi}{2} - y + k\pi$ — the 2-periodicity doubles the families. Reduce to one function first.

LESSON 10 / 10

Start from coordinates, not a sign mnemonic

On the unit circle, the point at angle θ is $P = (\cos\theta, \sin\theta)$. The x-coordinate gives cosine and the y-coordinate gives sine. Hence $\tan\theta = \sin\theta/\cos\theta$ whenever $\cos\theta \neq 0$. Quadrant signs follow immediately from the signs of the two coordinates.

One point contains all three basic ratios

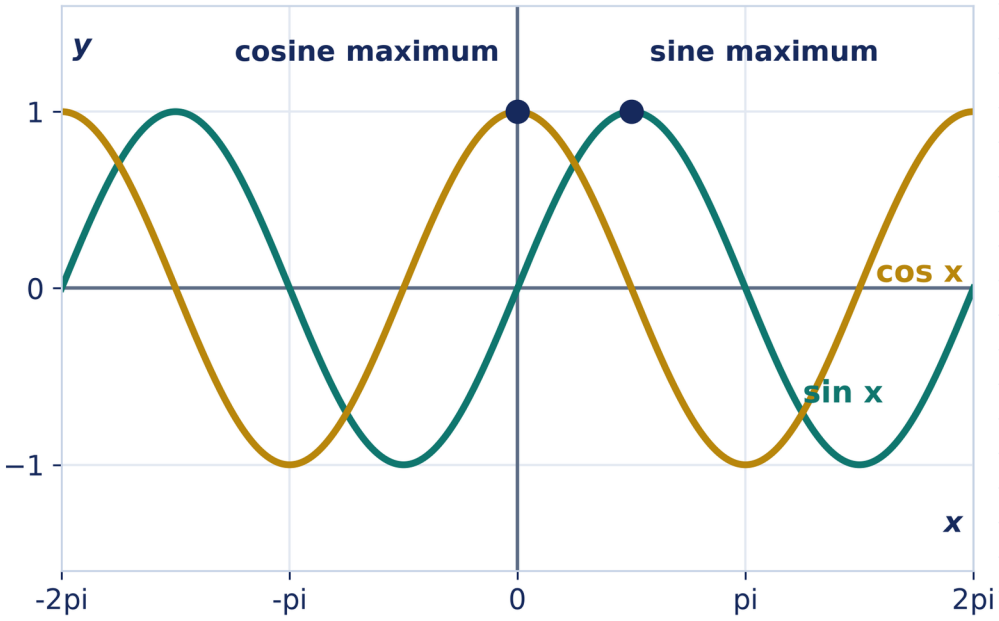


READ THE FIGURE
At 45 degrees the coordinates are equal and positive. Rotating the point changes signs and may interchange sine and cosine, but the coordinate definition never changes.

The radius also proves $\sin^2\theta + \cos^2\theta = 1$: it is simply the distance equation $OP = 1$. Dividing by $\cos^2\theta$ gives $1 + \tan^2\theta = \sec^2\theta$ where cosine is nonzero; dividing by $\sin^2\theta$ gives $1 + \cot^2\theta = \csc^2\theta$ where sine is nonzero. The excluded angles belong with the identities.

Sine and cosine are the same wave shifted

Read amplitude, period and phase from two curves



READ THE FIGURE

Both curves have amplitude 1 and period 2pi. The cosine curve shifted right by pi/2 becomes the sine curve: cos(x - pi/2) = sin x.

For $y = A\sin(Bx + C) + D$ with $A \neq 0$ and $B > 0$, the midline is $y = D$, amplitude is $|A|$, and period is $2\pi/B$. The horizontal shift is found by writing $Bx + C = B(x + C/B)$, so the shift is $-C/B$, **not** $-C$. Always read the coefficient of x after factoring B.

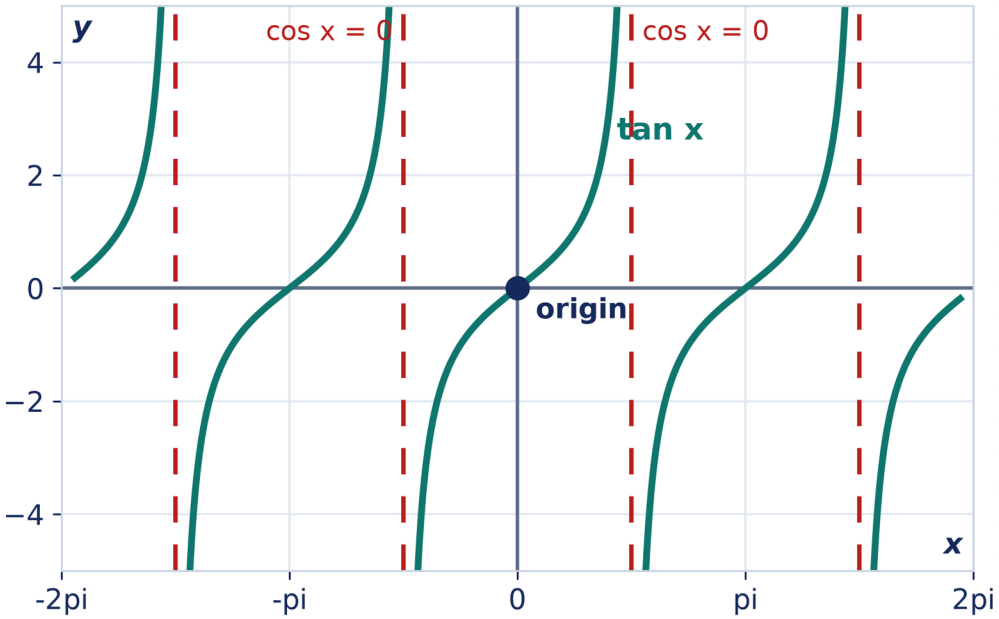
WORKED EXAMPLE

Transform before reading the period

For $y = 3\sin(2x - \pi/3) + 1$, write $2x - \pi/3 = 2(x - \pi/6)$. The midline is $y = 1$, amplitude 3, period π, and the basic sine wave is shifted right by $\pi/6$. Its maximum is 4 and minimum is -2. These values follow from the vertical stretch and shift; no point-by-point plotting is needed.

Tangent has a different period and breaks at asymptotes

Why tangent repeats after pi, not 2pi



READ THE FIGURE

Each branch is increasing on its domain. The dashed vertical lines are excluded inputs, not parts of the graph. Tangent has period pi because sin(x+pi) and cos(x+pi) both change sign.

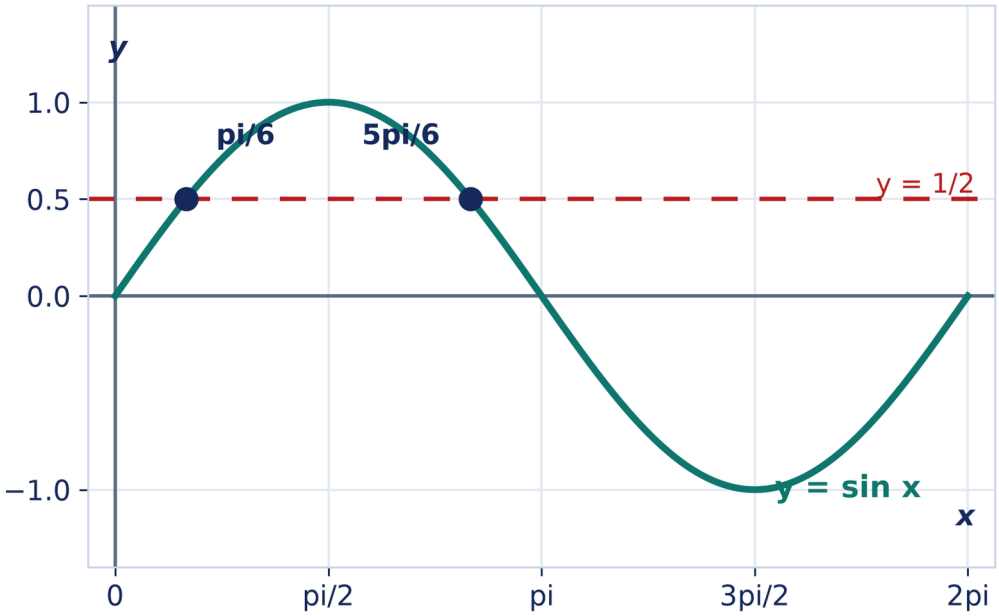
The statement "$\tan x$ is increasing" needs a domain qualification. It increases on every interval $(-\pi/2 + k\pi, \pi/2 + k\pi)$, but not across an asymptote. For example, $\tan(3\pi/4) = -1$ and $\tan(\pi/4) = 1$; comparing values across different branches does not preserve one global increasing order.

A horizontal line turns a trig equation into angles

To solve $\sin x = 1/2$, first find the intersections in one convenient interval. On $[0, 2\pi)$ they occur at $x = \pi/6$ and $x = 5\pi/6$. Then add $2n\pi$ because sine repeats after 2π.

Two intersections per period for sin x = 1/2

Base angles first, periodic copies second



READ THE FIGURE

The graph gives the two base angles. Periodicity supplies every other intersection; an interval in the question then selects only the required answers.

$$x = 2n\pi + \tfrac{\pi}{6} \quad \text{or} \quad x = 2n\pi + \tfrac{5\pi}{6}, \qquad n \in \mathbb{Z}.$$

COMMON TRAP

A general solution is not yet the requested answer

If the question asks for $x \in [0, \pi]$, substitute integer values of n or use the graph to retain only $\pi/6$ and $5\pi/6$. If a transformation introduced squaring, tangent division, or a denominator, also substitute into the original equation. "General solution" and "solution in the stated interval" are separate steps.

KEY RESULT

The graph-first equation routine

Reduce to one trigonometric function, draw one period, draw the required horizontal level, record all intersections in that period, and only then add the correct period. Use 2π for sine and cosine, and π for tangent. Finish with the interval and validity filters.

LESSON 01 / 04

Status of this sheet

The substitution $t = \tan\frac{A}{2}$ (Weierstrass) is **standard RMO/INMO technique** and a bridge to integral calculus; every model below is derivable from Part I identities. Nothing here "can never appear in JEE" — historically JEE-style papers have used it in disguised form; the classification is only that it exceeds the syllabus sentence "Trigonometric functions, general solution, identities".

The substitution and what it buys

Put $t = \tan\dfrac{A}{2}$. Then (for $A \neq \pi + 2k\pi$):

$$\sin A = \frac{2t}{1+t^2}, \qquad \cos A = \frac{1-t^2}{1+t^2}, \qquad \tan A = \frac{2t}{1-t^2}.$$

SHORT DERIVATION

Derivation — divide the double-angle forms by $\cos^2\frac{A}{2}$

$\sin A = 2\sin\frac{A}{2}\cos\frac{A}{2} = \frac{2\tan\frac{A}{2}}{\sec^2\frac{A}{2}} = \frac{2t}{1+t^2}$ (using $\sec^2 = 1+\tan^2$). Similarly $\cos A = \cos^2\frac{A}{2} - \sin^2\frac{A}{2} = \frac{1-t^2}{1+t^2}$ after dividing by $\cos^2\frac{A}{2}$. Tangent: ratio of the two ✓.

The point: **any rational expression in $\sin A, \cos A$ becomes a rational function of t** — trig equations become polynomial equations; trig integrals $\int R(\sin x, \cos x)\,dx$ become rational integrals (calculus toolkit).

WORKED EXAMPLE

Worked. Solve $\sin x + \cos x = 1$ for $0 \le x < 2\pi$ with $t = \tan\frac{x}{2}$

$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1 \Rightarrow 2t + 1 - t^2 = 1 + t^2 \Rightarrow 2t^2 - 2t = 0 \Rightarrow t = 0$ or $t = 1$. $t = 0 \Rightarrow x = 0$; $t = 1 \Rightarrow \frac{x}{2} = \frac{\pi}{4} \Rightarrow x = \frac{\pi}{2}$. The excluded point $x = \pi$ (where $\cos\frac{x}{2} = 0$) gives $\sin\pi + \cos\pi = -1 \neq 1$ ✓ nothing lost. Answer: $x \in \{0, \frac{\pi}{2}\}$ ✓.

COMMON TRAP

The blind spot — $A = \pi$ (mod 2π)

$t = \tan\frac{A}{2}$ is undefined when $\frac{A}{2} = \frac{\pi}{2} + k\pi$, i.e. $A = \pi + 2k\pi$. Every solution there is invisible to the substitution — **always test $A = \pi + 2k\pi$ in the original equation** and add it if it works. For $A \in (-\pi, \pi)$ the map $A \mapsto t$ is a bijection onto $\mathbb{R}$ (this is the clean domain window for the substitution).

KEY RESULT

When to use it. (1) Equations rational in $\sin, \cos$ that resist factorisation; (2) proving identities by clearing denominators in t; (3) converting geometry (slope $= \tan\frac{\theta}{2}$) into algebra; (4) (forward reference) any integral $\int R(\sin x, \cos x)\,dx$. Counter-indication: when half-angle or product-to-sum is quicker — do the cheap tool first.

Worked models

WORKED EXAMPLE

Worked. Solve $\sin x + \cos x = 1$ over all reals

General solutions from the two families: $x = 2n\pi$ (from $t = 0$) and $x = \frac{\pi}{2} + 2n\pi$ (from $t = 1$), $n \in \mathbb{Z}$; the blind-spot test $x = \pi + 2n\pi$ fails. Combined: $x = 2n\pi$ or $x = \frac{\pi}{2} + 2n\pi$ ✓ (leave as the two families).

WORKED EXAMPLE

Worked. Express $\frac{1-\cos x}{\sin x}$ in t

$\frac{1-\cos x}{\sin x} = \frac{1 - \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}} = \frac{\frac{2t^2}{1+t^2}}{\frac{2t}{1+t^2}} = t = \tan\frac{x}{2}$ ✓ — consistent with the half-angle identity of Part I. Any rational $R(\sin x, \cos x)$ now reduces by t-clearing denominators; $1 + t^2 \neq 0$ is automatic for real t.

Assumptions & conventions

A real with $A \neq \pi + 2k\pi$ under the substitution; the blind spot is checked separately in every problem; rational expressions in $\sin, \cos$ mean ratios of polynomials; all conversions verified back in the original equation.

LESSON 02 / 04

Status of this sheet

*Products like $\prod \sin \frac{k\pi}{n}$ are **classic RMO/INMO algebra** obtained by combining Part I identities with complex numbers (chapter 5 of the main sequence supplies $z = re^{i\theta}$ and $|z|$; only elementary modulus facts are used). Classification: beyond the syllabus sentence, but every step is derivable from supplied identities and the definition of $|1 - e^{i\theta}|$.*

The two flagship products

$$\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$$

$$\prod_{k=1}^{n} \cos \frac{k\pi}{2n+1} = \frac{1}{2^n}$$

SHORT DERIVATION

Derivation — factor $z^n - 1$ at the roots of unity, set $z = 1$, take moduli

The n roots of $z^n = 1$ are $e^{2\pi ik/n}$, $k = 0, \dots, n-1$. So $z^n - 1 = (z-1)\prod_{k=1}^{n-1}(z - e^{2\pi ik/n})$.

Cancel $z - 1$ ($\frac{z^n-1}{z-1} = 1 + z + \cdots + z^{n-1}$) and set $z = 1$: $n = \prod_{k=1}^{n-1}(1 - e^{2\pi ik/n})$. Now $|1 - e^{i\theta}| = 2\left|\sin \frac{\theta}{2}\right|$ (chord of a unit-circle arc; expand $|1 - \cos\theta - i\sin\theta|$). Taking moduli: $n = \prod 2\sin \frac{k\pi}{n} = 2^{n-1}\prod \sin \frac{k\pi}{n}$ (each $\sin \frac{k\pi}{n} > 0$ for $1 \le k \le n-1$) ✓. The cosine product comes from the same factorisation at $z = -1$.

WORKED EXAMPLE

Worked. $\prod_{k=1}^{6} \sin \frac{k\pi}{7}$ and $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$

$n = 7$: product $= \frac{7}{2^6} = \frac{7}{64} = 0.109375$ ✓ (direct: $0.43388 \times 0.78183 \times 0.97493 \times 0.97493 \times 0.78183 \times 0.43388 = 0.109375$).

Cosine product with $2n + 1 = 7$ ($n = 3$): $\frac{1}{2^3} = \frac{1}{8}$ ✓ (direct: $0.90097 \times 0.62349 \times 0.22252 = 0.125$).

The pattern to internalise

A product of trig values at rational multiples of π is a roots-of-unity problem in disguise. *Second tool: when the angles come from a polynomial whose roots are $\tan$-values or the values themselves, **Vieta** hands you sums/products for free (see the $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3}$ model of Part I sheet 8 — same idea with $\tan 3\theta$).*

WORKED EXAMPLE

Worked. Sine product for $n = 5$: $\sin 36^\circ \sin 72^\circ$?

$\prod_{k=1}^{4} \sin \frac{k\pi}{5} = \frac{5}{16}$. *Pair* $\sin \frac{\pi}{5} = \sin 36^\circ$, $\sin \frac{2\pi}{5} = \sin 72^\circ$, $\sin \frac{3\pi}{5} = \sin 72^\circ$, $\sin \frac{4\pi}{5} = \sin 36^\circ$: *product* $= (\sin 36^\circ \sin 72^\circ)^2 = \frac{5}{16}$, *so* $\sin 36^\circ \sin 72^\circ = \frac{\sqrt{5}}{4} \approx 0.5590$ ✓ *(direct:* $0.58779 \times 0.95106 = 0.55902$*)*.

COMMON TRAP

Traps on this sheet

- $\prod_{k=1}^{n-1}$ *vs* $\prod_{k=1}^{n}$*: the sine product has* $n - 1$ *factors (the* $k = 0$ *factor would be* $\sin 0 = 0$*); count factors before quoting.*
- $|1 - e^{i\theta}| = 2|\sin \frac{\theta}{2}|$ *— drop the absolute values only after checking positivity (done via* $0 < \frac{k\pi}{n} < \pi$*).*
- *The cosine product* $\prod_{k=1}^{n} \cos \frac{k\pi}{2n+1} = 2^{-n}$ *has* n *factors and denominator* $2n + 1$ *— a different index bookkeeping from the sine product.*
- *Claims here need complex arithmetic: state* $e^{i\theta} = \cos\theta + i\sin\theta$ *and* $|z|$ *facts explicitly in a solution write-up (assumptions below).*

Assumptions & conventions

Complex numbers used: $e^{i\theta} = \cos\theta + i\sin\theta$, $|zw| = |z||w|$, $|1 - e^{i\theta}| = 2|\sin \frac{\theta}{2}|$ *(proved above in modulus form). All products finite;* $n \geq 2$ *integer; positivity of* $\sin \frac{k\pi}{n}$ *for* $1 \leq k \leq n - 1$ *used to drop absolute values.*

LESSON 03 / 04

Status of this sheet

*The telescoping series $\sum \sin k\theta$ and the sums $\sum \cot^2$, $\sum \csc^2$ are **standard olympiad algebra**; all derive from Part I transformation formulae and the roots-of-Vieta technique of sheet 2. Models are derivable from supplied information; classified as extensions because they sit beyond the current syllabus sentence.*

Sums with arguments in arithmetic progression (telescoping)

The trick: *multiply by $2\sin\frac{\theta}{2}$ and watch the product-to-sum formula telescope.*

$$\sum_{k=1}^{n} \sin k\theta = \frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

$$\sum_{k=1}^{n} \cos k\theta = \frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

SHORT DERIVATION

Derivation — the $2\sin\frac{\theta}{2}$ multiplier

$2\sin\frac{\theta}{2}\sin k\theta = \cos\left(k-\frac{1}{2}\right)\theta - \cos\left(k+\frac{1}{2}\right)\theta$. *Summing over $k = 1 \ldots n$ telescopes to* $\cos\frac{\theta}{2} - \cos\left(n+\frac{1}{2}\right)\theta$. *Then apply* $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$ *with* $C = \frac{\theta}{2}$, $D = \left(n+\frac{1}{2}\right)\theta$; $= 2\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}$. *Divide by* $2\sin\frac{\theta}{2}$ ✓. *(Valid for* $\sin\frac{\theta}{2} \neq 0$, *i.e.* $\theta \notin 2\pi\mathbb{Z}$; *the excluded cases are computed directly as sums of 0/1.)*

WORKED EXAMPLE

Worked check. $n = 3$, $\theta = \frac{\pi}{6}$

Direct: $\sin 30^\circ + \sin 60^\circ + \sin 90^\circ = 0.5 + 0.86603 + 1 = 2.36603$. *Formula:* $\frac{\sin 45^\circ \sin 60^\circ}{\sin 15^\circ} = \frac{0.70711 \times 0.86603}{0.25882} = 2.36603$ ✓.

Sums of squares at rational multiples of π (roots + Vieta)

The numbers $t_k = \tan\frac{k\pi}{n}$, $k = 1, \ldots, n-1$, are the non-zero roots of a polynomial obtained from $\tan(n\theta)$ in terms of $t = \tan\theta$ (write $\tan n\theta = \frac{\Im(1+it)^n}{\Re(1+it)^n}$ and set the numerator to 0). Vieta then gives sums of powers.

$$\sum_{k=1}^{n-1} \cot^2\frac{k\pi}{n} = \frac{(n-1)(n-2)}{3}, \qquad \sum_{k=1}^{n-1} \csc^2\frac{k\pi}{n} = \frac{n^2-1}{3}.$$

SHORT DERIVATION

Derivation sketch — cotangents as roots of the complementary polynomial

$\cot^2\frac{k\pi}{n} = \frac{1}{t_k^2}$, and the symmetric identity $\sum\frac{1}{t_k^2} = \left(\sum\frac{1}{t_k}\right)^2 - 2\sum_{j<k}\frac{1}{t_j t_k}$ turns Vieta's data for the t_k into the closed form $\frac{(n-1)(n-2)}{3}$. Then $\csc^2 = 1 + \cot^2$ adds $(n-1)$ terms of 1: $\sum\csc^2 = (n-1) + \frac{(n-1)(n-2)}{3} = \frac{(n-1)(n+1)}{3} = \frac{n^2-1}{3}$ ✓.

WORKED EXAMPLE

Worked checks. $n = 3, 4, 5$

$\cot^2$: $n=3$: $\cot^2 60^\circ + \cot^2 120^\circ = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} = \frac{2\cdot 1}{3}$ ✓. $n=4$: $1 + 0 + 1 = 2 = \frac{3\cdot 2}{3}$ ✓. $n=5$: $4 = \frac{4\cdot 3}{3}$ ✓. $\csc^2$: $n=3$: $\frac{4}{3} + \frac{4}{3} = \frac{8}{3} = \frac{9-1}{3}$ ✓; $n=4$: $2 + 1 + 2 = 5 = \frac{15}{3}$ ✓; $n=5$: $8 = \frac{24}{3}$ ✓.

KEY RESULT

Whenever a sum runs over trig values at equally spaced angles: (1) look for a telescoping multiplier ($2\sin\frac{\theta}{2}$ for AP-argument sums); (2) look for the polynomial whose roots they are (Vieta for square/power sums); (3) pair symmetric terms ($k \leftrightarrow n-k$) to halve the work; (4) validate a small case numerically before the proof.

COMMON TRAP

Traps on this sheet

- The telescoping formulas divide by $\sin\frac{\theta}{2}$: state $\theta \notin 2\pi\mathbb{Z}$ (or handle those cases directly).
- $\sum\sin k\theta$ closed form: the numerator is $\sin\frac{n\theta}{2}\cdot\sin\frac{(n+1)\theta}{2}$ — order $(n+1)$ vs n swapped between the two factors vs the cosine version (which has $\cos\frac{(n+1)\theta}{2}$).
- $\cot^2\frac{k\pi}{n}$ has a term at $k = \frac{n}{2}$ (when n even) equal to 0 — include it (it is $\cot^2\frac{\pi}{2} = 0$).
- $\tan n\theta$'s polynomial in $t = \tan\theta$ has **degree** n in the tangent-angle formula bookkeeping; only the roots t_k with $\tan$ defined count.

Assumptions & conventions

$n \geq 2$ integer; θ real with $\sin\frac{\theta}{2} \neq 0$ in series formulas; Vieta applied to real polynomials; numeric checks above computed in radians (converted to degrees for readability).

LESSON 04 / 04

Status of this sheet

Important classification note. The current JEE Advanced syllabus sentence for trigonometry is "Trigonometric functions. General solution of trigonometric equations. Identities." — it does **not** list properties of triangles or heights & distances. The formulae below are the **standard triangle toolkit** used across JEE Main-level papers, olympiads (RMO/INMO geometry-algebra), and later chapters (conics, 3D, vectors). They are added theory (not derivable from the identities alone — they encode triangle geometry), but every model built here only uses Part I identities plus the sine/cosine rules as stated. Do not assume they "can never appear"; assume they are simply outside the quoted syllabus sentence and learn them as toolkit.

The four triangle formulae

For a triangle with sides a, b, c opposite angles A, B, C, circumradius R, inradius r, semiperimeter $s = \frac{a+b+c}{2}$, area Δ:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \qquad a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}bc\sin A = \sqrt{s(s-a)(s-b)(s-c)}, \qquad r = \frac{\Delta}{s}$$

$$\tan\frac{A}{2} = \frac{r}{s-a}$$

SHORT DERIVATION

Cosine rule and $\Delta = rs$ — one line each

Cosine rule: drop a height $h = c\sin B$ from C; project: $a = b\cos C + c\cos B$-style coordinate computation gives $a^2 = b^2 + c^2 - 2bc\cos A$ (or expand $|\vec{B} - \vec{C}|^2$ in the vectors chapter). $\Delta = rs$: split the triangle into three (r, s)-shaped pieces — $\Delta = \frac{1}{2}ra + \frac{1}{2}rb + \frac{1}{2}rc = rs$ ✓. $\tan\frac{A}{2} = \frac{r}{s-a}$: the incircle tangent lengths are $s-a, s-b, s-c$ and half-angle right triangles give the ratio.

Derived special values (pure identities, pentagon angles)

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4} \approx 0.30902$$

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4} \approx 0.80902$$

SHORT DERIVATION

Derivation — double-angle, then quadratic

Let $x = 18^\circ$, so $5x = 90^\circ$ and $2x = 90^\circ - 3x$, hence $\sin 2x = \cos 3x$. Expand: $2\sin x\cos x = 4\cos^3 x - 3\cos x$; with $\cos x \neq 0$: $2\sin x = 4\cos^2 x - 3 = 4(1 - \sin^2 x) - 3 = 1 - 4\sin^2 x$. So $4\sin^2 x + 2\sin x - 1 = 0 \Rightarrow \sin x = \frac{-2+\sqrt{4+16}}{8} = \frac{\sqrt{5}-1}{4}$ (positive root; $\sin 18^\circ > 0$) ✓. Then $\cos 36^\circ = 1 - 2\sin^2 18^\circ = \frac{\sqrt{5}+1}{4}$ ✓ (check numerically: 0.809017).

WORKED EXAMPLE

Worked. Triangle with $A = 60^\circ$, $b = 5$, $c = 8$; find a and Δ

Cosine rule: $a^2 = 25 + 64 - 2 \cdot 5 \cdot 8 \cdot \frac{1}{2} = 49 \Rightarrow a = 7$. $\Delta = \frac{1}{2} \cdot 5 \cdot 8 \cdot \frac{\sqrt{3}}{2} = 10\sqrt{3} \approx 17.32$. Cross-check with Heron: $s = \frac{7+5+8}{2} = 10$, $\sqrt{10 \cdot 3 \cdot 5 \cdot 2} = \sqrt{300} = 10\sqrt{3}$ ✓.

WORKED EXAMPLE

Worked. R and r for the same triangle

$2R = \frac{a}{\sin A} = \frac{7}{\sqrt{3}/2} = \frac{14}{\sqrt{3}} \Rightarrow R = \frac{7}{\sqrt{3}} \approx 4.041$. $r = \frac{\Delta}{s} = \frac{10\sqrt{3}}{10} = \sqrt{3} \approx 1.732$. Check $\tan\frac{A}{2} = \tan 30^\circ = \frac{1}{\sqrt{3}}$ vs $\frac{r}{s-a} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$ ✓.

COMMON TRAP

Traps — triangle toolkit

- The sine rule gives an **ambiguous case** (SSA): $a = b\frac{\sin A}{\sin B}$ with $\sin B = \frac{b\sin A}{a}$ may admit two triangles (B and $180^\circ - B$).
- $\Delta = \frac{1}{2}$(base)(height) needs the height; $\frac{1}{2}bc\sin A$ needs the angle **between** the two sides.
- Heron's formula: s is semiperimeter; writing $s = a + b + c$ is the classic slip.
- $\tan\frac{A}{2} = \frac{r}{s-a}$ requires the usual triangle convention ($s > a$); for degenerate/failing configurations state the inequality.
- 18°-family values ($\sin 18^\circ$, $\cos 36^\circ$, $\sin 54^\circ = \cos 36^\circ$ etc.): derive from the quadratic each time if memory fails — and pick the sign from the quadrant (18° is QI).

Assumptions & conventions

Triangle = non-degenerate (positive angles summing to π, positive sides satisfying triangle inequalities); R circumradius, r inradius, s semiperimeter, Δ area; special values derived from double-angle identities as shown; toolkit status per the note at the top of this sheet.